

Philippe Heymans Smith

Data & AI SME

Staff Data Scientist and AI Subject Matter Expert with a proven track record of architecting large-scale machine learning pipelines and multi-agent AI systems. Adept at scaling cross-functional teams and translating complex AI capabilities into accessible, high-impact business solutions in healthcare and finance applications.

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Experience

Growth Acceleration Partners Remote
Staff Data Scientist — Data & AI SME Aug 2025 – Present

- Directed the architectural design and operational excellence of a large-scale, high-stakes healthcare propensity modeling pipeline, scaling a core group of 3 Data Scientists into a robust, cross-functional team of 15 Data Scientists, Data Engineers, and MLOps professionals.
- Engineered a multi-stage LangGraph AI agent for healthcare audience design with multiple parallel and adversarial validators while compressing project delivery timelines by 10–15x against baseline estimates.
- Engineered deterministic grounding strategies for domain-sensitive healthcare data to minimize LLM hallucinations, integrating FAISS vector similarity, regex matching, and Pydantic schema validation to ensure absolute accuracy for ICD-10 and NDC codes.
- Advised enterprise-wide technical strategy as a core member of the AI Council and an AI Garage Judge, consulting on strategic AI implementation, mentoring project leads, and advising on ML pipelines for complex client engagements.
- Implemented adversarial auto-validation loops utilizing multi-agent AI frameworks, deploying independent generator and validator agents to rigorously stress-test code deliveries, prevent architectural drift, and identify edge cases prior to human sign-off.
- Transformed highly technical machine learning capabilities into accessible, interactive stakeholder tools by rapidly prototyping complex proof-of-concepts UIs, accelerating cross-departmental alignment and Product team buy-in.
- Upskilled cross-functional teams in AI-fluent development practices, advanced data science concepts, data pipelines, solution architecture, and healthcare compliance pipelines, scaling execution capacity well beyond our linear headcount growth.

Growth Acceleration Partners Remote
Senior Data Scientist — Data Science and AI SME Jul 2022 – Aug 2025

- Project lead of client’s healthcare modelling initiative.
- Designed and maintained two iterations of an end-to-end modelling tool: the first AWS-native leveraging SageMaker and Athena; the second on Snowflake with an H2O.ai backend coordinated by an internally developed Python API client.
- Contributed to GAP’s AI and cloud strategy, published articles on GenAI ethics, and delivered talks on responsible ML development.

Growth Acceleration Partners San José, Costa Rica
Advanced Data Scientist Apr 2021 – Jul 2022

- Developed scoring models for audience design, including research for medical target definitions using ICD standard codification.
- Built a dynamic SQL querying tool that automated query building for all target definition approaches.
- Established collaboration channels with universities and women-led STEM organizations through the Analytics Center of Excellence.

Mastercard SJO Center of Excellence San José, Costa Rica
Associate Consultant, Advanced Analytics Jul 2019 – Apr 2021

- Delivered advanced analytics products for financial institutions across Latin America, including segmentations, campaign diagnostics, product migration models, and fraud diagnostics.
- Developed two proprietary in-house R packages with 84+ generalized functions, making advanced R capabilities accessible across the team.
- Designed and delivered R training programs for non-specialist teams in Brazil, México, Colombia, Uruguay, Argentina, and Costa Rica.

Quanto Soluciones Analíticas y Cuantitativas S.A. San José, Costa Rica
Financial Analyst, Mathematical Modeling Department Apr 2018 – Jul 2019

- Rewrote the R calibration algorithm for Liquidity Risk indicators’ sensitivity against macroeconomic indexes.
- Spearheaded the delivery of 10 additional features for a proprietary Liquidity Risk analysis tool for the second largest bank in Costa Rica.

Certifications

Neo4j Certified Professional Mar 2023
Neo4j

Microsoft Certified: Azure Fundamentals Nov 2022
Microsoft

AWS Certified Solutions Architect – Associate Mar 2022
Amazon Web Services (AWS)
Credential ID: DDPQJ13DGGJ11Q89T

Practical Machine Learning on H2O Jul 2020
Coursera
Credential ID: 9VEDDWAYKPPD

The R Programming Environment Jul 2020
Coursera
Credential ID: CVC7VKML283X

Education

Data ScienceTech Institute Applied MSc
Data Science & Artificial Intelligence 2020 – 2023

Universidad de Costa Rica, School of Mathematics Bachelor of Science
Actuarial Science 2013 – 2018

Skills

Machine Learning Predictive modeling, Propensity modeling, Clustering, Time series, Model validation	Generative AI & LLMs Multi-agent systems, Prompt engineering, RAG, Hallucination mitigation
Solution Architecture System design, API design, Workflow orchestration, Proof-of-concept development	Cloud & MLOps AWS SageMaker, Snowflake, Athena, Production ML pipelines
Technical Leadership Team scaling, AI strategy, Mentorship, Stakeholder communication	Healthcare AI Audience design, ICD-10/NDC compliance, Regulatory awareness, Clinical data pipelines

Open Source

{chronicle} 2020 – Present
Author and maintainer of {chronicle}, an R package for creating frictionless R Markdown reports with a sequential syntax inspired by keras/ggplot2. Published to CRAN on January 21st, 2021. Presented at ConectaR 2021 and useR! Conference 2021.